MILITARY SPECIFICATION

UNDECYLENIC ACID SOLUTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Undecylenic Acid Solution (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a clear solution containing the following ingredients on a w/w basis:

Undecylenic Acid	10.0 percent
n-Propyl Alcohol	40.0 percent
Propylene Glycol	20.0 percent
Triethanolamine	2.0 percent or sufficient to adjust pH
Suitable Perfume	0.18 percent
Polysorbate 80	1.0 percent
Purified Water	qs 100 percent

3.1.1 pH. The pH of the solution when determined potentiometrically at 25°C, using the USP method, shall be between 6.0 to 7.0 when the solution is diluted one part solution to nine parts of purified water.

3.1.2 Specific gravity. The specific gravity of the solution when determined at 25°C using an pycnometer or specific gravity balance, shall be not less than 0.9230 and not more than 0.9330.

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3.1.3 Presenece of oily droplets. The solution shall not show the presence of oily droplets when tested as specified in 4.7.1.

3.1.4 Assay of undecylenic acid. The solution shall assay to contain not less than 9.50 percent and not more than 10.50 percent of undecylenic acid when asayed as specified in 4.7.2.

3.2 Ingredients.

3.2.1 Polysorbate 80, propylene glycol, purified water, triethanolamine and undecylenic acid. The polysorbate 80, propylene glycol, purified water, triethanolamine and undecylenic acid used in the manufacture of the solution shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the manufacture of the solution shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto; or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.4 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the solution to the date of delivery to the Government.

3.5 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of the inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

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4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate containers
Minor	II	2.5	Filled immediate containers

4.3.1 End item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
pH	3.1.1	S-2	5 } Composite
Specific gravity	3.1.2	S-2	5 } Composite
Presence of oily droplets	3.1.3	S-2	5 } Composite
Assay of undecylenic acid	3.1.4	S-2	5 } Composite

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Solution not free from foreign matter.
102	Solution has crystal formation.
103	Solution not homogeneous.
104	Solution not clear.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

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4.5 Packaging inspection. The inspection of the immediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Presence of oily droplets.

Dilute 10 ml of the finished solution with 90 ml of purified water and mix thoroughly by shaking for 2 minutes.

4.7.2 Assay for undecylenic acid.

Weigh approximately 1.5 grams of finished solution to the nearest milligram. Transfer the solution quantitatively to a 500 ml, separatory funnel. Add 200 ml of purified water and acidify with 5 ml of concentrated hydrochloric acid. Mix thoroughly and extract with two, 100 ml portions of petroleum ether (BP 30° - 65°C). Add sodium chloride, as necessary, to obtain clear separations. Let stand from 15 minutes to one (1) hour (or as necessary) to permit emulsion to break. Discard aqueous layer. Wash the combined petroleum ether extracts with purified water until free of mineral acid. Evaporate the combined acid free, petroleum ether extracts to approximately 5 ml on a steam bath, add approximately 100 ml of neutralized specially denatured alcohol (SD 3A) and titrate with 0.1N sodium hydroxide using phenolphthalein as the indicator.

Calculation:

$$\frac{(\text{ml of NaOH}) \times (\text{Normality of NaOH}) \times 0.1843 \times 100}{\text{sample weight (grams)}} = \text{percent of undecylenic acid}$$

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle, containing fifty-nine ml, as specified, constitutes one unit of issue.

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5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be commercially available light resistant bottle considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in substantial commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

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5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. In addition, the date of manufacture shall be shown in lieu of the date packed. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-237-8468	UNDECYLENIC ACID SOLUTION, 59 ml

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 6505-1910

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-U-37763 UNDECYLENIC ACID SOLUTION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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